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Implementation of SUBNETTING in Cisco packet tracer simulator.

Aim:

To implement subnetting in Cisco packet tracer, design multiple subnet with a suitable number of hosts, assign static IP address & establish communication b/w different subnets using a router.

Introduction:

Subnetting is the process of dividing a larger network into smaller logical networks called subnets. Each subnet can contain a suitable number of hosts. subnetting helps in organizing networks and allow different groups of hosts to communicate through a router.

In this experiment, multiple subnets are designed using Cisco packet tracer. static IP address are assigned to the hosts and router interfaces. the different subnets are then connected through a router and packet transmission is observed

Requirements:

* Cisco packet tracer
* Pcs / Hosts
* Network Connection
* Router
* Suitable IP address

Procedure :

Step 1: Create the network.

i) open cisco packet tracer.

ii) place the required number of PCs/hosts in the work

iii) place a router to connect the different subnets.

iv) divide the hosts into multiple groups based on the required subnet design.

v) connect the host belonging to each subnet.

Step 2: Design multiple subnets.

i) create a multiple subnets according to the requi numbers of hosts

ii) select a suitable IP network for subnet

iii) divide the network into smaller subnetwork

iv) ensure the each subnet has sufficient IP addr for its hosts

v) Identify the network address and suitable ho address for each subnet.

Step 3:

Connect the host belonging to each subn using the required network connections

Step 4:

Assign suitable static IP address and s masks to all the hosts

Step 5: Configure the appropriate default gatewa for each host

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Step 6: Connect the different subnets using a router

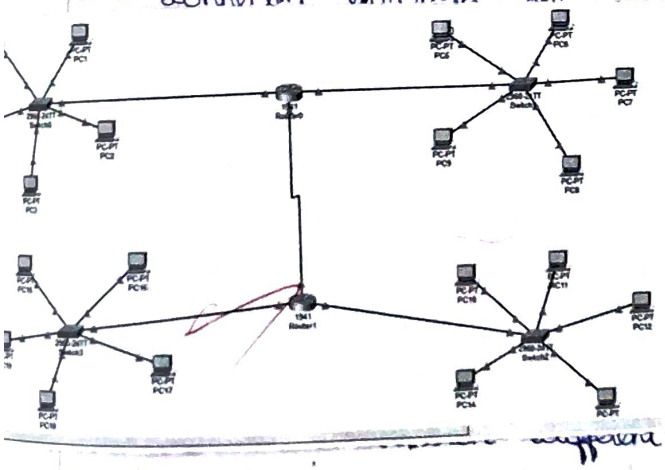
Step 7: Configure the router interfaces with suitable IP address and subnet masks.

Step 8: Check the IP configuration and connectivity of all the devices.

Step 9: Perform packet transmission between hosts in the same subnet and different subnets.

Step 10: Use simulation mode to observe the packet travelling from one subnet to another through the router.

Step 11: Test packet communication when the subnet are connected through the router.



without connecting router and compare

is required for

subnet.

Results:

Thus, the multiple subnets were successfully designed and implemented using cisco packet tracer and communication between different subnet was verified